Night-Time Traffic Light Recognition Based on Enhancement-Guided Object Detection

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***Abstract*—Traffic light recognition is crucial for autonomous driving. While significant progress has been made in favorable conditions, recognition performance in night-time scenes remains a challenge. One straightforward approach is to apply enhancement methods that improve degraded images prior to object detection. However, since most enhancement methods are tailored for human perception; they may not consistently improve recognition accuracy for machine learning techniques. To address this, we propose an enhancement-guided framework for night-time traffic light recognition, called EG-TLR. EG-TLR consists of a residual denoising module (RDM) and a mixed attention traffic light detection module (MATLDM). The RDM reduces noise in degraded night-time images while preserving essential traffic light features by extracting sparsity information and performing context aggregation. The MATLDM improves feature extraction and recognition performance in complex night-time scenes by incorporating a shadow detection layer (SDL) and a mixed attention module (MAM). Moreover, to address the lack of a dedicated night-time traffic light dataset, we construct the Night-TL dataset utilizing publicly available images. Extensive experiments on Night-TL and LISA datasets demonstrate that EG-TLR achieves an AP50 of 79.83% and an AP50:95 of 36.62%, with an inference speed of 5.9 ms and 16.9 GFLOPs, outperforming other state-of-the-art methods. Furthermore, ablation studies and visualization results validate the effectiveness of our proposed method. The Night-TL dataset can be downloaded from: https://github.com/feiqinaqian/Night-TL-dataset.**

***Index Terms*—Deep learning, intelligent driving, night-time image processing, traffic light recognition, traffic safety.**

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I. Introduction

TRAFFIC light intersections are among the most congested and dangerous areas in road traffic, characterized by the convergence of motor vehicles, nonmotor vehicles, and pedestrians. These intersections are more susceptible to traffic accidents than is commonly acknowledged [1], [2]. To enhance safety, intelligent transportation systems (ITS), intelligent vehicles (IV), and autonomous driving systems (ADS) have been vigorously developed in recent years [3], [4], [5]. The accuracy and robustness of traffic light recognition are crucial for these systems in making informed decisions and ensuring safe navigation [6]. Unlike general object detection, traffic light recognition requires extraction on both color (e.g., green and red signals) and structure features (e.g., black rectangular box). Due to their small size and distant appearance in the field of view, effective recognition methods must capture and preserve fine-grained details. Furthermore, as traffic light recognition is integral to real-time decision-making in intelligent driving systems, it demands high-speed processing to enable timely responses and ensure operational safety. Over the past years, many vision-based traffic light recognition methods have been proposed [7], [8], [9]. While these methods perform well in favorable conditions, they face significant challenges in adverse environments, particularly at night.

Fig. 1(a) illustrates real-world night-time traffic light scenes, where reduced visibility blurs the features of traffic lights, adding to the environmental complexity. The self-luminous nature of traffic lights, which are brighter than their surroundings, often result in detail loss due to overexposure, often obscuring their shape and edges. Additionally, interference from other light sources, such as streetlights, advertising lights, and vehicle taillights, further obscures the distinctive features of traffic lights. As a result, the degraded features, low contrast between traffic lights and their backgrounds, and the overall complexity of the environment make night-time traffic light recognition particularly challenging, significantly impacting the reliability of autonomous driving under all conditions. Fig. 1(b) illustrates recognition failures of YOLOv8-x under

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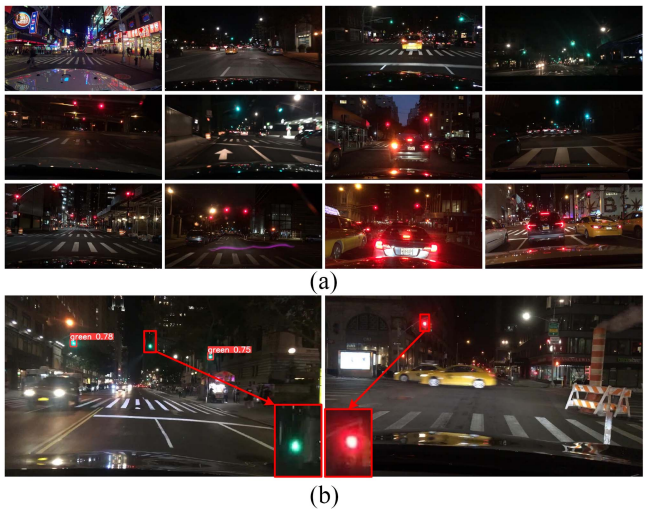


Fig. 1. (a) Illustration of complex night-time traffic light scenes. It includes unclear features, overexposures, and interference from other light sources in various environments. (b) Recognition failures by YOLOv8-x under night-time conditions.

these conditions. Noise and interference obscure features, such as black rectangular box, and although color signals still remain, the loss of structural context leads to recognition failures. The degraded structural and color information leads to miss recognition, emphasizing the necessity of image enhancement to strengthen recognition frameworks and ensure robust performance.

A straightforward solution for addressing visually degraded traffic light images is to enhance the brightness and visibility through preprocessing with image enhancement methods. These methods have been shown to enhance human visual perception and various low-level vision tasks [10], [11]. However, when these enhanced images are used as input for recognition methods, the results are not consistently positive, and in some cases, performance may even degrade [12]. This degradation occurs because enhancement modules are primarily designed to enhance the human viewing experience rather than specifically enhancing traffic light recognition.

One possible solution is to integrate the low-light enhancement and detection modules within a unified framework, optimizing both simultaneously. This solution logically suggests that enhanced visibility in night-time traffic light images can improve overall performance. However, these methods often introduce significant levels of noise amplification and color distortion [13], which may further degrade the traffic light features. Additionally, because traffic lights are self-luminous, applying low-light enhancement methods can cause unnatural changes in contrast between the traffic light and the background. As a result, the expected performance improvement is limited and may even result in a decline. Furthermore, because traffic lights are small and their features are more degraded at night than during the day, current detection modules struggle with complex backgrounds and low visibility, resulting in unsatisfactory performance. Therefore, this raises a critical research question: How can we design unified and effective enhancement and detection modules to achieve high performance in night-time traffic light recognition?

To address the above issues, we propose an enhancement-guided traffic light recognition method, called EG-TLR, specifically designed for night-time conditions. EG-TLR comprises of two key components: a residual denoising module (RDM) and a mixed attention traffic light detection module (MATLDM). Unlike common low-light enhancement methods, the RDM focuses on reducing the noise interference while preserving key features of night-time traffic lights, achieving noise removal without requiring referenced images. Additionally, the MATLDM is introduced to further improve feature extraction and enhance recognition performance under challenging night-time conditions. The RDM and MATLDM are integrated within a unified framework and jointly optimized using a shared detection loss, ensuring that the enhancements by the RDM directly benefit the MATLDM. This approach enables the MATLDM to leverage enriched features from the RDM, achieving superior performance and providing a robust solution for night-time traffic light recognition.

To facilitate evaluation and address the absence of datasets for night-time traffic light recognition, we utilize public images and construct a new dataset called Night-TL. The main contribution in this study can be summarized as follows.

1) We propose EG-TLR, an enhancement-guided object detection method designed specifically for night-time traffic light recognition. This method integrates an enhancement module and a detection module, enabling robust performance in challenging night-time conditions.
2) We develop the RDM as the enhancement module, which effectively removes noise from night-time traffic light images while preserving critical features. Furthermore, we incorporate MATLDM, which enhances feature extraction and improves recognition accuracy. Experimental results demonstrate that EG-TLR outperforms other state-of-the-art methods.
3) We construct the Night-TL dataset, the first dataset tailored for night-time traffic light recognition, which provides a valuable resource for advancing research in traffic light recognition under adverse conditions.

The remainder of this article is organized as follows. Section II provides a comprehensive review of related work. In Section III, we describe in detail the proposed EG-TLR method. Section IV introduces the Night-TL dataset for night-time traffic light recognition. Section V presents the experiments and evaluations of our method. Finally, Section VI concludes the article and suggests future research directions.

II. Related Works

Night-time traffic light recognition involves both low-level vision tasks e.g., low-light enhancement and denoising) and high-level vision tasks e.g., semantic segmentation and object detection), which are closely related to three key research areas, i.e., traffic light recognition, night-time image enhancement, and enhancement-guided object detection in night-time scenes.

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A. Traffic Light Recognition

Traffic light recognition aims at accurately recognizing the status of light signals and helping IVs navigate safely at intersections. Existing methods can be classified into two groups: 1) empirical-based methods and 2) deep learning-based methods.

In the early stages, research works on traffic light recognition mainly relied on empirical-based methods. These methods often involved color thresholding, shape analysis, and template matching performed by domain experts to identify the traffic lights [14], [15], [16], [17]. However, due to the large number of hyperparameters, these methods are not robust and prone to being affected in varying and complex scenarios, which significantly limit their applications [18]. With the rapid advancement of computer vision, deep learning-based methods have become prevalent in traffic light recognition. Unlike empirical-based methods, deep learning-based methods can learn abstract feature representations directly from image [19], [20]. Li et al. [21] used weighted feature integration, dynamic IOU threshold adjustment, and dynamic parameter selection in the loss function to improve traffic light recognition. Zhang et al. [22] introduced a self-adaptive preprocessing module for hazy and low-light images, and a two-stage Yolov8-based model to recognize traffic light colors and shapes using a new dataset. Wang and Zhou [23] introduced a real-time traffic light recognition model using dual-channel images. A color saliency-based model was proposed to distinguish colors in the dark channel, and a CNN-based classifier was utilized to reduce false recognition in the bright channel. Ouyang et al. [24] proposed a CNN-based multibackbone network to locate potential traffic light areas, followed by a color space classifier for class determination. Yao et al. [25] presented a traffic light recognition method using a coarse-to-fine approach, with an enhanced super class detector for location determination followed by a fine-grained classifier for class distinction. Golcarenarenji et al. [26] integrated a modified Yolov3 model on an edge-powered platform to achieve fast traffic light recognition. Liu et al. [27] employed a capsule neural network for scene understanding and traffic light signals recognition.

Although these methods have achieved impressive performance of traffic light recognition, most of them face significant challenges in night-time scenes, resulting in a substantial drop in performance.

B. Night-Time Image Enhancement

The goal of night-time image enhancement is to improve the visibility of information obscured or disrupted in the dark areas of an image, which has gained significant attention recently [28], [29]. Due to the inherently low visibility and inevitable noise of night-time images, enhancement methods mainly include the low-light enhancement and denoising methods.

1) Low Light Image Enhancement: The goal of low-light enhancement methods is to improve the interpretability of images captured in night environment. Li et al. [30] presented a progressive neural network adopting a dual attention model and a residual block to improve image brightness and contrast. Liang et al. [10] enhanced regional brightness consistency using unpaired images and a bag of curves to generate negative samples. Ma et al. [31] introduced a self-calibrated learning framework employing a cascaded illumination learning process with a self-calibrated module for convergence. Li et al. [11] proposed a nonreferenced curve estimation method to dynamically adjust pixel values by incorporating a lightweight deep network. Cai et al. [32] adopted a retinex-based transformer network with an illumination guided mechanism to enhance low-light images in a one-stage manner.

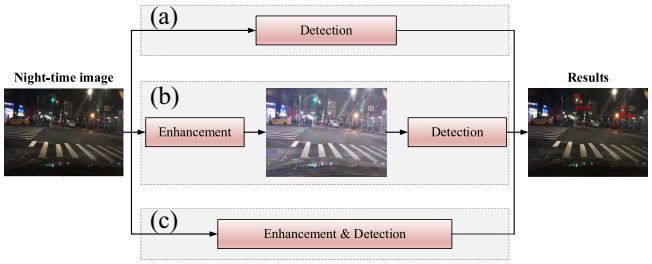


Fig. 2. Comparison of three approaches to object detection with enhancement module in night-time scenes. (a) Directly training of object detection module. (b) Separately training of the enhancement and object detection modules. (c) Unified training the enhancement and object detection modules.

2) Image Denoising: Image denoising aims to extract accurate structural information and remove noise, thereby generating clean images. Tian et al. [33] presented a CNN-based denoising method incorporating wavelet transforms. Zhang et al. [34] introduced a tunable noise map to guide CNN for denoising and accurate context information acquisition. Li et al. [35] combined gradient direction and edge-preserving regularization to enhance image denoising. Pan et al. [36] employed an iterative correction scheme and a guided feature domain denoising residual network to predict and correct noise levels. Huang and Dragotti [37] adopted lifting-inspired invertible neural networks for multiscale noise removal and a noise estimation network to adjust denoising strength.

When enhancement methods are applied to traffic light recognition, the improvements are not always positive. These methods are primarily designed to enhance human visual perception rather than recognition accuracy, which can lead to potential conflicts.

C. Enhancement-Guided Object Detection in Night-Time Scenes

Object detection in night-time scenes is crucial for robust perception. Many CNN-based detectors, such as YOLO [38], SSD [39], and DeTR [40], have been proposed. These methods have attracted considerable scholarly interest, with successful applications in various areas [41], [42]. However, their performance often struggles under night-time conditions. Generally, there are three approaches to integrating image enhancement and object detection methods to improve the performance [43], as illustrated in Fig. 2: (a) *Direct Approach:* The object detection module is trained directly on night-time images; (b) *Separate Approach:* First, the enhancement module is trained to generate enhanced night-time images.

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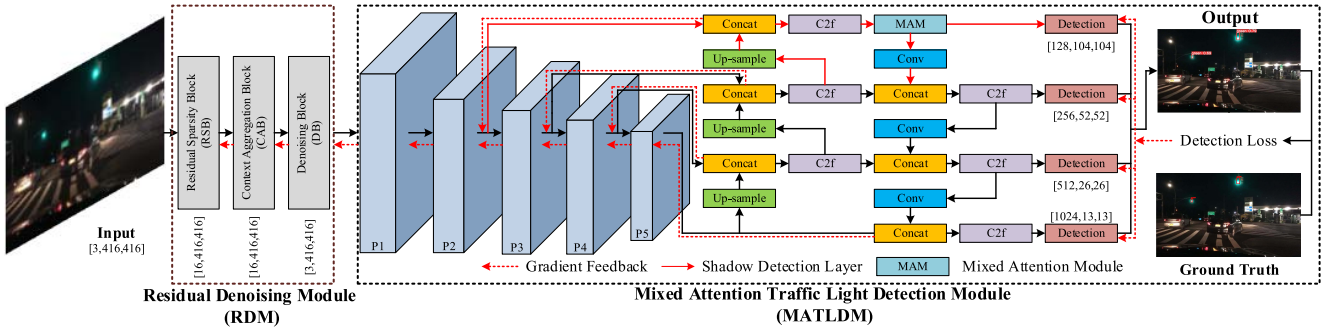


Fig. 3. Pipeline of the proposed EG-TLR. The RDM predicts and removes noise from night-time traffic light images, while the MATLDM enhances detection capabilities. Both modules are trained together in an end-to-end manner. The dotted red line indicates gradient feedback.

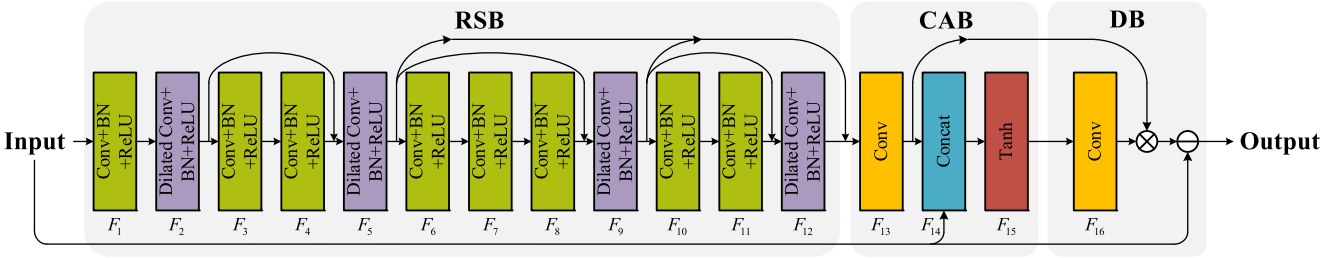


Fig. 4. Details of RDM, composed of three submodules: the RSB, the CAB, and the DB.

Then, the detection module is trained on these enhanced images; (c) *Unified Approach:* The enhancement and object detection modules are trained together in an end-to-end manner.

Detectors trained using the direct approach often fail to achieve satisfactory performance, hindered by the low contrast and degradation common in night-time scenes [44]. In the separate approach, the enhancement and detection modules are trained independently. However, since the objectives of enhancement and detection differ, potential conflicts can arise. Wang et al. [45] suggested that the enhancement methods might suppress detector performance by increasing background interference. Few studies have paid attention to the unified approach, where enhancement and detection modules are jointly trained [46], [47], [48], [49]. Liu et al. [46] integrated image enhancement and object detection modules in parallel, employing a feature guided module to connect the two, thereby enhancing detection performance under various scenes. Liang et al. [47] combined recurrent exposure generation with multiexposure detection modules to address nonuniform illumination and noise in night-time images, thus enhancing detection performance. Liu et al. [48] utilized a differentiable image processing module with hyperparameters predicted by a small CNN-based parameter predictor, improving image enhancement and detection under adverse weather conditions. Yin et al. [49] proposed a pyramid enhancement network integrated with YOLOv3 to improve object detection visibility in low-light conditions.

Unlike these methods, we address the night-time traffic light recognition challenges with a unified network combining noise suppression and an enhanced detection module. These modules collaboratively reduce noise, enhance features, and achieve precise recognition, significantly improving recognition performance in night-time conditions.

III. Methodology

In this section, we provide an overview of our proposed EG-TLR framework for the joint task of enhancement and detection, followed by a detailed description of the method.

A. Overview

The proposed EG-TLR employs an enhancement-guided mechanism to improve the accuracy of night-time traffic light recognition. As illustrated in Fig. 3, the workflow of EG-TLR consists of two key modules: the RDM and the MATLDM. The RDM estimates and removes noise while enhancing feature representation in night-time traffic light images. The MATLDM employs a mixed attention module (MAM) and a shadow detection layer (SDL) to capture detailed features of traffic lights, thereby further enhancing recognition performance. The entire network operates in an end-to-end manner and is trained with detection loss to achieve enhancement-guided night-time traffic light recognition. The dotted red lines in Fig. 3 represent the gradient feedback throughout the entire EG-TLR.

B. Residual Denoising Module

Many existing denoising methods require paired noise and noise-free images for training. However, obtaining reference noise-free images in traffic light recognition tasks is challenging. Additionally, most denoising methods rely on deep CNNs to improve performance, which places a significant burden on the denoising module and complicates deployment in enhancement-guided night-time traffic light recognition. To address these limitations, we proposed the RDM, which can effectively predict and remove noise features from traffic light images without the need for reference noise-free images. As demonstrated in Fig. 4, the RDM comprises three submodules:

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the residual sparsity block (RSB), the context aggregation block (CAB), and the denoising block (DB).

1) RSB: Sparse information exploits signal sparsity to enable efficient noise suppression by isolating essential features from random noise, a property that is particularly critical for high-performance CNN-based denoising [50], [51]. Meanwhile, contextual features capture spatial relationships and hierarchical dependencies among pixels, aiding the model in understanding semantic connections within an image.

To reduce network complexity while effectively leveraging both sparse information and contextual features, we incorporate dilated convolution into the RSB to improve denoising ability. The architecture of RSB consists of 12 CNN layers, including two types: dilated convolution layer and standard convolution layer. Each is composed of a stacked Dilated Conv+BN+ReLU and Conv+BN+ReLU block. The dilated convolution can expand the receptive field, enabling the network to capture sparse information effectively. However, an excessive number of dilated convolution layers can impose a significant computational cost, four dilated convolution blocks, are strategically placed at second, fifth, ninth, and twelfth layers. A dilated factor of 2 is selected to effectively capture sparse information while preserving computational efficiency. Unlike feature extraction architectures, such as DenseNet [52], which use dense connectivity with concatenation, each dilated convolution layer in RSB is connected to the previous one in a residual manner via addition. The number of channels in the first layer matches that of the input image, while the others are set to 16. The combination of dilated and standard convolution layers promotes sparsity, enabling the capture of more useful information and ensuring more stable convergence.

Assume that the input night-time traffic light image is denoted as I_N, the intermediate features of the ith layer as F_i, and the output of the RSB as O_R. The formulation of the RSB can be expressed as

$$O_R = D(R(F_9, S(S(D(R(F_5, S(S(S(D(R(F_2, S(S(D (S(I_N)))))))))))))))) \quad (1)$$

where D represents a dilated convolution layer, S represents a standard convolution layer, and $R(,)$ is the residual connection.

2) CAB: After the input image is mapped into a higher dimensional space by the RSB, the useful global and local features may weaken. To better leverage these features, we propose CAB to perform effective context aggregation through multipath connections, as shown in Fig. 4. Specifically, multiple contexts from the previous two dilated convolution layers in the RSB are fused, and then passed through a Conv layer with a kernel size of $16\times3\times3\times3$. Finally, the output feature of the Conv layer and the input image are concatenated to enhance context aggregation. Additionally, a Tanh function is adopted to normalize the feature maps. This process is explained as follows:

$$O_C = T(\mathrm{Cat}(C(F_5, F_9, F_{12}))) \quad (2)$$

where T, Cat, and C represent the Tanh, concatenation, and convolution functions, respectively. O_C is the output of CAB, and (F_5, F_9, F_{12}) represents the context aggregation paths from previous layers.

3) DB: The intricate background of night-time scenes can obscure features, making them blend in and thereby increasing the difficulty of training [50]. To help the network capture and remove key noise features in night-time traffic light images, the DB is designed to use information from the current stage to guide previous stage. Specifically, a 1×1 Conv layer with the filter size of $6\times3\times3\times3$, matching the input image channels, is utilized to compress features from the F_{15} into adaptive feature weighting. Subsequently, this adaptive feature weighting is multiplied with features from F_{13} to preserve prominent noise features. The implementation process can be expressed with the following formula:

$$O_m = C(F_{15}) \times F_{13} \quad (3)$$

where O_m represents the multiplied features and can be understood as the predicted noise features. By learning to predict the noise (rather than directly learning the denoised image), the model can be optimized more effectively.

Finally, the learned noise features are removed from the input image I_N to reconstruct the clean images. The final output O_{RDM} can be expressed as

$$O_{\mathrm{RDM}} = I_N - O_m. \quad (4)$$

C. Mixed Attention Traffic Light Detection Module

YOLOv8 is widely deployed in various real-world applications. This work adopts YOLOv8-n as the detection module for EG-TLR. To further enhance the night-time traffic light recognition performance while maintaining low computational complexity, we propose the MATLDM, as shown in Fig. 3. This module integrates the SDL and the MAM.

1) SDL: Traffic lights often appear smaller compared to other objects, particularly when viewed from a distance. While deeper layers of the network primarily capture abstract representations, shallow layers are crucial for preserving structure details required for accurate small object detection. To addresses the challenges of small night-time traffic light recognition, we propose an efficient SDL by focusing on fine-grained feature extraction from the shadow layers of the detection module, as the red line in Fig. 3.

Specifically, the feature maps from the shadow layer of P_2 are concatenated with the up-sampled features from P_3, and then passed through a C2f module [53] and a MAM module. Subsequently, the feature maps are divided into two branches. One branch is fused with deeper layers via a convolution layer to fully utilize the features from the shadow layer. The other branch is used for detection with a resolution of $128\times104\times104$. The feature maps from different layers of P_2, P_3, P_4, and P_5 used for detection can be expressed as

$$O_F = \mathrm{Det}(P_2, P_3, P_4, P_5) \quad (5)$$

where O_F is the final output of MATLDM, Det() represents the detection layer.

2) MAM: Spatial features, including the salience information of objects at different locations, are crucial for determining the night-time traffic light location. In the SDL, as the image undergoes several convolutional layers with down-sampling operations, key features are

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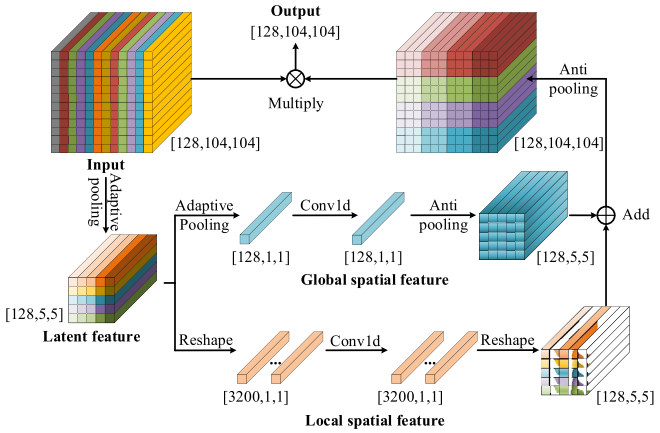


Fig. 5. Illustration of MAM, which integrates the spatial features into overall feature representation.

extracted, but their spatial information is lost, leading to diminished detection performance. Moreover, current attention mechanisms primarily focus on channel information while often overlooking spatial features [54]. To effectively retain both the channel information and spatial information, we employ a MAM mechanism to leverage mixed feature representation in SDL, as illustrated in Fig. 5.

First, the input feature maps from the SDL undergo an adaptive pooling operation to generate latent features with a resolution of $128\times5\times5$. To comprehensively capture spatial information, two branches are developed to transform the latent feature into one-dimensional (1-D) vectors based on the initial pooling, capturing both global and local spatial information. One branch performs adaptive pooling to generate global spatial features with a resolution of $128\times1\times1$, while the other branch generates local spatial features through a reshaping operation with a resolution of $3200\times1\times1$. Both branches are then passed through 1-D convolutions (Conv1d) to extract relationships and dependencies between different channels. The size of Conv1d kernel k is proportional to the channels, which means that only the interaction between each channel and its k adjacent channels is considered when capturing local cross-channel interaction information. The determination of k is determined as follows:

$$k = \left| \frac{\log_2 C}{\alpha} + \frac{\beta}{\alpha} \right|_{\text{odd}} \tag{6}$$

where C denotes the number of channels, α and β are hyperparameters set to 2, and odd indicates that k must an odd number; otherwise, it is incremented by 1.

Subsequently, the resolution of two features is restored to $128\times5\times5$, and they are added together to fuse the features, thereby leveraging mixed feature representation. Finally, an anti-pooling layer focuses on the primarily features, scales them to the original size of input, and multiplies them with the input to highlight important features and generate final output.

D. Loss Function

The loss function of EG-TLR aligns with the original YOLOv8, consisting of three subcomponents: the object bounding box loss L_{bou} [55], the object classification loss L_{cls}, and the object distribution focal loss L_{dis} [56]. The total loss L_{total} can be described as follows:

$$L_{\text{total}} = \lambda_1 L_{\text{bou}} + \lambda_2 L_{\text{cls}} + \lambda_3 L_{\text{dis}} \tag{7}$$

where λ_1, λ_2, and λ_3 are the corresponding weights of three sections, set to 7.5, 0.5, and 1.5, respectively.

IV. Night-TL Dataset

Data plays a key role in the task of night-time traffic light recognition. Currently, there is a lack of benchmark datasets specifically designed for this task. To mitigate this gap, we construct the night-time traffic light (Night-TL) dataset by gathering two real-world datasets: 1) the BDD100K dataset [57] and 2) the NightCity dataset [58].

A. Dataset Collection

The BDD100K dataset, developed to advance autonomous driving research, is one of the largest and most diverse driving video datasets. It comprises one hundred thousand driving videos captured at different times of the day, various weather conditions, and diverse geographic locations. The NightCity dataset is specifically curated for research on night-time urban driving scenarios and includes data from cities, such as Los Angeles, New York, Hong Kong, and London. We manually select a total of 4382 images from these two datasets to ensure a comprehensive representation of various night-time traffic light scenarios.

B. Processing and Statistics

Due to the challenges of low illumination, varying light sources, and the small size of night-time traffic light in the original datasets, many labels are inaccurate or missing. To ensure accurate annotations, we employ the "LabelImg" tool for relabeling. This process is conducted in two steps by two annotators. First, each annotator performs an initial check and labeling of their assigned portion of the images. Next, they exchange their annotated images for cross-verification. Any discrepancies between the annotations are required to be discussed and resolved through consensus, ensuring both quality and consistency in the labeling process. The final annotations are stored in XML file following the PASCAL VOC format [56].

To achieve robust training and evaluation, data augmentation techniques, such as horizontal flipping are applied, resulting in a total of 7369 night-time traffic light images. The Night-TL dataset includes two classes of traffic lights: "green" and "red". 70% images are allocated for training, while the remaining 30% are reserved for testing. Table I presents detailed statistics of the Night-TL dataset. Fig. 6 provides visual examples from the dataset.

V. Experiments

In this section, we first present the implementation details and evaluation metrics. Next, we describe the experimental results in comparison with the state-of-the-art methods.

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TABLE I
STATISTICS OF THE NIGHT-TL DATASET

	Categories	Numbers	Annotations	
			Green	Red
Set	Train	14217	9802	4415
	Test	6162	4288	1874
Total	Overall	20379	14090	6289

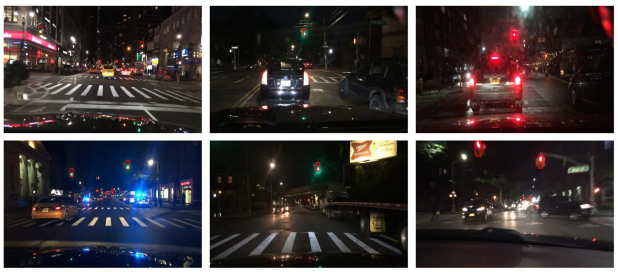


Fig. 6. Samples of Night-TL dataset.

Finally, we conduct an ablation study to demonstrate the effectiveness of the proposed methods.

A. Experimental Setting and Evaluation Metrics

The experiments are conducted on Ubuntu 22.04 LTS system utilizing four Nvidia GeForce RTX 4090 GPUs with 24 GB of memory, and an Intel Xeon Platinum 8352V CPU. Training is performed from scratch, without pretrained weights, using the SGD optimizer with an initial learning rate of $1e^{-2}$ and a decay of $5e^{-4}$. The batch size is set to 64, and training is carried out for a total of 300 epochs. The early stopping strategy is employed to prevent overfitting. Considering the higher computational demands at larger input dimensions, the input images of our experiments are set to a resolution of 416×416 for all experiments.

To validate the effectiveness of the proposed method, we use the mean average precision (AP50) as the evaluation metric, calculated as the average of precision values across all classes at an IOU threshold of 0.5. This process is defined as follows:

$$\text{AP50} = \frac{1}{m}\int_0^1 P(R)dR \tag{8}$$

where m denotes the number of classes, P and R denote precision and recall, respectively, defined as

$$P = \frac{\text{TP}}{\text{TP}+\text{FP}} \tag{9}$$

$$R = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{10}$$

where TP is the number of true predicted positive instances, FP is the number of false predicted positive instances, and FN is the number of false predicted negative instances. Additionally, AP50:95 is also used for evaluation, representing the average AP50 value across IOU thresholds from 0.5 to 0.95, incremented by 0.05.

Furthermore, GFLOPs and inference time (ms) are utilized to assess the computational efficiency and speed of the compared methods, providing a comprehensive evaluation of both accuracy and performance.

B. Comparison With State-of-the-Art (SOTA) Methods

As introduced in Section II-C, enhancement-guided object detection methods are grouped into three categories: 1) the direct approach; 2) the separate approach; and 3) the unified approach. We compare the performance of our proposed method with SOTA methods in each category to demonstrate its effectiveness. Furthermore, to comprehensively evaluate its effectiveness, we extended our evaluation on the night sequences of LISA traffic light dataset, referred to as LISA-Night. The results are presented in Table II.

1) Comparison With Directly Trained Approaches: We evaluate the proposed EG-TLR against six directly trained recognizers: 1) YOLOv8-n [59]; 2) YOLOv8-x [59]; 3) YOLOv9-c [60]; and 4) YOLOv10-x [61]; 5) YOLOv11-x [62]; and 6) RT-DETR-R101 [63].

Table II indicates that RT-DETR-R101 achieves the highest AP50 of 73.09% within the direct approach category, reflecting its superior performance. However, this comes with the relatively high computational cost, at 191.0 GFLOPs. YOLOv8-n is the most lightweight model with 8.1 GFLOPs, but achieves a lower AP50 of 64.60%. YOLOv10-x and YOLOv11-x strike a balance between accuracy and computational efficiency, with AP50 of 65.53% and 70.42%, respectively. Notably, our proposed EG-TLR outperforms all directly trained models by a margin of 6.74% over the best direct approach method.

2) Comparison With Separately Trained Approaches: To compare with advanced separately trained methods, we adopt the low-light enhancement methods PIE [10] and Zero-DCE [11] to enhance the night-time traffic light images, followed by YOLOv8-n for recognition.

As shown in Table II, Zero-DCE and PIE achieve an AP50 of 63.78% and 57.08%, respectively. In contrast, the proposed EG-TLR outperforms these methods by 16.05% and 22.05%, respectively. Notably, compared to direct trained methods, the performance of Zero-DCE and PIE decreases by 0.82% and 7.52%, respectively. This suggests that the separate approach method does not effectively improve night-time traffic light recognition performance on Night-TL dataset. A possible explanation is that these methods introduce additional noise that hinder performance.

3) Comparison With Unified Trained Approaches: We compare our proposed EG-TLR with four unified-trained methods: SCINet [31], Retinexformer [32], IAT [64], and IA-YOLO [48]. SCINet employs a self-calibrated module and an illumination estimation module with shared weights to enhance night-time images. Since only the illumination estimation module is used during testing, we utilize it exclusively for experiments. Retinexformer estimates illumination and utilizes an illumination-guided transformer to model long-range dependencies and restore image quality. IAT features a local branch for pixel-wise adjustments and a global branch to estimate image signal processor parameters. IA-YOLO incorporates a differentiable image processing module, with hyperparameters adaptively predicted by a small CNN-based

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TABLE II
PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS ON NIGHT-TL DATASET AND LISA-NIGHT DATASET

Method	GFLOPs	Time [ms] RTX 4090	Time [ms] Jetson Orin	Night-TL [AP50%]			LISA-Night [AP50%]						
				Green	Red	Average	Go	Stop	Warn	StopL	GoL	WarnL	Average
Direct Approach													
YOLOv8-n [59]	**8.1**	**4.0**	**29.8**	68.44	60.75	64.60	76.61	84.53	75.97	66.85	88.83	90.61	80.57
YOLOv8-x [59]	257.4	11.8	130.5	74.28	65.35	69.81	77.96	87.54	83.42	72.22	93.19	99.50	85.64
YOLOv9-c [60]	102.1	8.8	80.8	72.14	63.90	68.02	78.42	87.28	78.00	71.19	91.80	99.50	84.36
YOLOv10-x [61]	160.4	10.3	122.3	69.04	62.03	65.53	71.99	83.16	78.88	65.36	92.82	87.83	80.01
YOLOv11-x [62]	194.4	12.1	116.8	74.44	66.39	70.42	78.29	87.98	81.34	72.62	93.43	99.50	85.53
RT-DETR -R101 [63]	191.0	27.5	241.5	76.98	69.19	73.09	92.16	94.22	91.47	**89.85**	95.37	57.44	86.75
Separate Approach													
PIE [10] +YOLOv8-n	93.6	5.1	41.8	59.66	54.50	57.08	79.16	87.27	80.20	70.90	93.64	92.00	83.86
Zero-DCE [11] + YOLOv8-n	8.2	5.0	40.1	67.76	59.81	63.78	78.56	87.58	79.12	71.48	93.47	99.50	84.95
Unified Approach													
SCINet [31]	8.5	4.2	34.0	69.58	60.45	65.02	76.99	85.04	74.60	67.50	90.47	90.61	80.87
Retinexformer [32]	52.2	13.1	202.9	69.30	61.01	65.16	77.33	85.12	76.92	69.27	90.85	85.06	80.76
IAT [64]	11.9	10.8	173.8	65.09	55.34	60.22	78.68	87.23	81.37	73.79	93.88	99.50	85.74
IA-YOLO [48]	109.7	24.9	220.5	66.84	62.60	64.72	77.89	86.34	79.05	68.89	90.21	93.10	82.58
EG-TLR (ours)	16.9	5.9	65.7	**81.73**	**77.92**	**79.83**	**93.76**	**95.34**	**92.76**	88.51	**98.63**	**99.50**	**94.75**

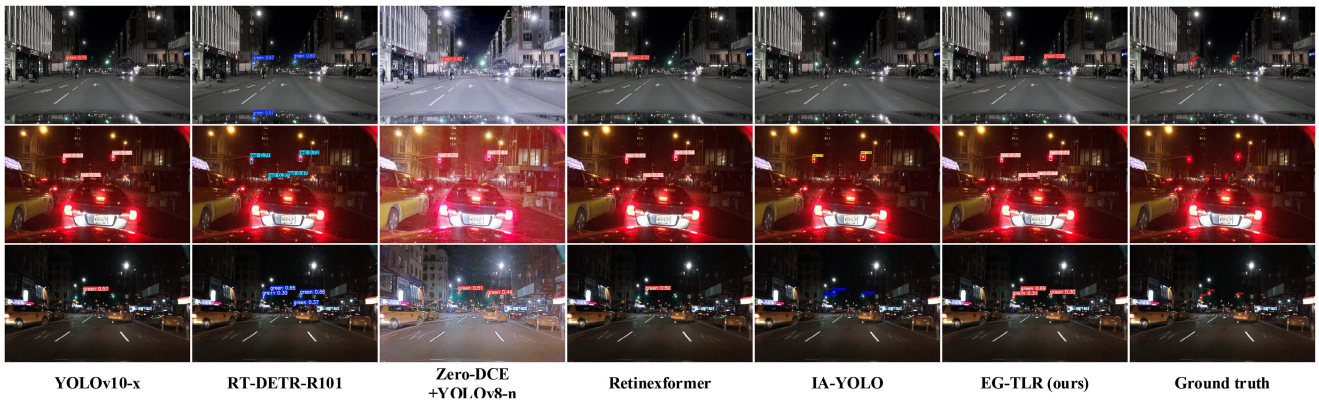


Fig. 7. Qualitative visualization of our proposed EG-TLR compared with SOTA methods (YOLOv10-x, RT-DETR-R101, Zero-DCE+YOLOv8-n, Retinexformer, and IA-YOLO) on the Night-TL dataset.

parameter predictor, enabling the model to enhance images and improve detection performance. As shown in Table II, EG-TLR demonstrates superior performance compared to the other unified-trained methods. Specifically, EG-TLR achieves an average AP50 of 79.83%, which is 14.81%, 14.67%, 19.61%, and 15.11% higher than SCINet, Retinexformer, IAT, and IA-YOLO, respectively. The improvements in recognition accuracy can be attributed to the effectiveness of both enhancement and detection module, which together enhance night-time traffic light recognition performance.

Fig. 7 visualizes the recognition results of EG-TLR compared with other SOTA methods, including the direct, separate, and unified approaches. The results demonstrate that existing SOTA methods struggle with night-time traffic light recognition; for instance, many methods encounter missed recognitions or false recognitions across different scenarios, and Retinexformer even falsely recognizes the advertising light as a red traffic light. In contrast, the proposed EG-TLR significantly outperforms these methods, effectively handling unclear environments, lost traffic light features, and interference from other light sources. This comparison underscores the effectiveness and robustness of EG-TLR in night-time traffic light recognition tasks.

4) Evaluation on LISA-Night Dataset: To comprehensively validate the efficacy of our approach, we extend the evaluation to the LISA Traffic Light Dataset, specifically focusing on the nighttime sequences (referred to as LISA-Night) to align with the scope of our research on traffic light recognition under night-time conditions. The comparative results are summarized in Table II.

The proposed EG-TLR achieves superior performance across the majority of traffic light categories, including “Go”, “Stop”, “Warn”, “StopL”, “GoL”, and “WarnL”, attaining the highest average AP50 of 94.75%. This represents an improvement of 8.00% over the second-best method, RT-DETR-R101, which achieves 86.75%. As illustrated in Fig. 8, EG-TLR effectively addresses the challenges of nighttime conditions, reducing false positives and missed detections compared to RT-DETR-R101. These findings highlight the robustness

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Fig. 8. Qualitative visualization of our proposed EG-TLR compared with RT-DETR-R101 on the LISA-Night dataset.

and effectiveness of EG-TLR in handling complex nighttime environments. Moreover, the consistent superior performance across both the Night-TL and LISA-Night datasets underscores its generalizability and suitability for night-time traffic light recognition tasks.

5) Performance Comparison on Inference Time: To evaluate the inference efficiency of our proposed EG-TLR model, we compared its performance with state-of-the-art methods using two hardware platforms: 1) NVIDIA RTX 4090 and 2) Jetson Orin. No optimization techniques, such as TensorRT, are applied during the evaluations.

On the high-performance GPU NVIDIA RTX 4090, EG-TLR achieved an inference time of 5.9 ms, surpassing computationally intensive models, including YOLOv11-x with 12.1 ms, RT-DETR-R101 with 27.5 ms, and IA-YOLO with 24.9 ms. On the embedded Jetson Orin platform, EG-TLR achieved an inference time of 65.7 ms, significantly outperforming models, such as YOLOv11 with 116.8 ms, RT-DETR-R101 with 241.5 ms, and Retinexformer with 202.9 ms.

The results indicate that EG-TLR offers a practical and efficient solution for real-world applications, particularly in resource-constrained environments like autonomous vehicles and embedded systems. This balance between computational cost and accuracy demonstrates the suitability of our model for deployment in complex nighttime traffic scenarios.

6) Comparison With Different Enhancement Modules: To evaluate the effectiveness of proposed RDM, we use MATLDM as the baseline and integrate it with SCINet, Retinexformer, IAT, and RDM, respectively. To further analyze the impact of network architectures in RDM, we integrate MATLDM with a 14-layers standard CNN that matches the number of convolutional layers in RDM. The standard CNN follows the same Conv+BN+ReLU structure as the standard convolution used in the RSB.

As shown in Table III, the baseline without any enhancement module achieves an AP of 76.89% and an AP50:95 of 34.28%. Incorporating a 14-layer CNN yields limited improvements, with an AP50 of 77.93% and an AP50:95 of 34.64%, indicating that deeper networks alone provide limited performance improvements under night-time conditions. Integration of SCINet slightly decreases performance, yielding an AP50 of 76.83% and an AP50:95 of 33.99%. Retinexformer improves performance, raising the AP50 to 78.21% and the AP50:95 to 35.43%, thereby outperforming SCINet. IAT provides modest gains over the baseline, with an AP of 77.15% and an AP50:95 of 33.50%. These results indicate that improvements from low-light enhancement methods are limited and can sometimes degrade recognition performance. In contrast, our proposed RDM achieves the highest performance, with an AP50 of 79.83% and an AP50:95 of 36.62%, marking significant improvements of 1.62% in AP50 and 1.19% higher in AP50:95 over the second-best results. Fig. 9 illustrates the recognition results of different enhancement modules integrated into MATLDM. In the first row, which contains four traffic lights, most modules fail to recognize all traffic lights, except for the baseline and our RDM. In the second row, SCINet, Retinexformer, and our RDM modules successfully recognize all traffic lights, while the baseline and IAT modules fail. In the third row, only the baseline and Retinexformer fail to recognize all the traffic lights. Overall, these results demonstrate that our proposed RDM achieves the best performance across these scenarios.

TABLE III
EFFECTIVENESS OF DIFFERENT ENHANCEMENT MODULES INTEGRATED INTO THE MATLDM

Method	AP50			AP50:95
	Green	Red	Average	
Baseline	79.71	74.07	76.89	34.28
CNN	80.38	75.48	77.93	34.64
SCINet [31]	80.40	73.26	76.83	33.99
Retinexformer [32]	80.68	75.75	78.21	35.43
IAT [64]	81.06	73.23	77.15	33.50
RDM (ours)	**81.73**	**77.92**	**79.83**	**36.62**

TABLE IV
ABLATION STUDIES OF THE PROPOSED EG-TLR

RDM			MATLDM		AP50	AP50:95
RSB	CAB	DB	MAM	SDL		
					64.60	27.09
✓					66.25	27.90
	✓				65.12	27.29
		✓			64.79	27.19
✓	✓				65.99	28.27
✓	✓	✓			66.58	28.54
✓	✓	✓	✓		67.16	28.68
✓	✓	✓		✓	78.85	35.89
✓	✓	✓	✓	✓	**79.83**	**36.62**

C. Ablation Study

To evaluate the effectiveness of each component in EG-TLR, we conduct ablation studies using the Night-TL dataset.

1) Effectiveness of RDM: Using YOLOv8-n as the baseline, we independently add the CAB, RSB, and DB components of the RDM to evaluate each component's contribution. As Table IV shows, each component contributes positively to the overall recognition performance. Specifically, the addition of RSB results in the most significant improvement, increasing the AP50 by 1.65%, and the AP50:95 by 0.81%. CAB and DB yield smaller gains, with AP50 increases of 0.52% and 0.19%, and AP50:95 increases of

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Fig. 9. Visualization results of different enhancement modules integrated into the MATLDM.

TABLE V
IMPACT OF DILATED CONVOLUTION IN RDM

Convolutions in RSB	AP50	AP50:95
Standard convolution	78.85	35.89
Dilated Convolution	**79.83**	**36.62**

TABLE VI
IMPACT OF INTEGRATING MAM INTO DIFFERENT LAYERS IN MATLDM

P_2	P_3	P_4	P_5	AP50	AP50:95
				78.85	35.89
✓				**79.83**	36.62
✓	✓			79.46	**36.77**
✓	✓	✓		79.25	36.31
✓	✓	✓	✓	78.34	35.83



Fig. 10. Visualization of ground truth images and their corresponding heatmaps, with and without RDM.

0.20% and 0.10%, respectively. When DB module is removed from the RDM while retaining other components, performance decreases by 0.59% in AP50 and 0.27% in AP50:95, demonstrating its role in stabilizing feature quality and enhancing robustness under night-time conditions. When three components are integrated, they achieve the highest improvements in both AP50 and AP50:95, demonstrating their combined impact on enhancing performance.

To assess the effectiveness of dilated convolutions in the RSB, we replace them with standard convolutions, and the results are presented in Table V. The findings indicate that this replacement leads to a performance decline, with AP50 dropping from 79.83% to 78.85% and AP50:95 decreasing from 36.62% to 35.89%. These results underscore the importance of incorporating dilated convolutions in the RSB to enhance overall recognition performance.

2) Effectiveness of MATLDM: To evaluate the effectiveness of MATLDM, we separately remove the SDL and MAM from EG-TLR. The results show that the inclusion of the MAM increases the AP50 by 0.58% and AP50:95 by 0.14%. In contrast, replacing MAM with alternative mechanisms, such as ECA-Net [65] resulted in a performance decline, with AP50 and AP50:95 dropping to 79.12% and 35.92%, respectively. The addition of SDL significantly improves performance, increasing AP50 by 12.27% and the AP50:95 by 7.35%. When both MAM and SDL are combined in MATLDM, the AP50 and AP50:95 reach 79.83% and 36.62%, respectively, indicating that both components effectively enhance overall performance.

Table VI illustrates the performance impact of integrating MAM into different layers within MATLDM. Integrating MAM solely into P_2 layer achieves the relatively high performance, achieving an AP50 of 79.83% and an AP50:95 of 36.62%. In contrast, applying MAM to other layers, such as P_4 and P_5, leads to a decline in accuracy. This reduction in performance may be attributed to the decreased sensitivity of deeper layers to the fine-grained spatial and channel features required for detecting small traffic lights.

3) Visualization of Heatmaps: Fig. 10 illustrates the heatmaps of the detection module with and without RDM, along with the corresponding ground truth images. The results show that the feature maps with RDM exhibit stronger activation values compared to those without RDM, demonstrating that the proposed RDM enables the recognition method to more precisely focus on potential traffic lights in night-time conditions.

4) Effect of Recognition Results With the Proposed RDM and MATLDM: To assess the effect of RDM and MATTLDM on recognition performance, we use YOLOv8-n as the baseline

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Fig. 11. Recognition results using RDM and MATLDM.

TABLE VII
DIFFERENT FEATURE-LEVEL INTEGRATED INTO THE MATLDM

RDM Components	AP50	AP50:95	Time
RSB	78.29	35.45	5.3
CAB	77.82	34.33	**4.4**
RSB + CAB	79.26	36.01	5.8
RSB + CAB + DB	**79.83**	**36.62**	5.9

TABLE VIII
IMPACT OF DIFFERENT INPUT IMAGE SIZE ON THE PROPOSED EG-TLR

Input Size	AP50			AP50:95
	Green	Red	Average	
320×320	76.84	71.03	73.93	32.46
416×416	81.73	77.92	79.83	36.62
640×640	**83.68**	**79.91**	**81.80**	**39.70**

and visualize the results in Fig. 11. In the first row, both the baseline and RDM fail to recognize two traffic lights at a long distance, while RDM+MATLDM successfully recognizes all traffic lights. In the second row, the baseline misses one traffic light. Although RDM and RDM+MATLDM successfully recognize all traffic lights, RDM+MATLDM achieves higher confidence scores. These results indicate that the proposed EG-TLR effectively enhances overall performance.

5) Effect of Recognition Results at Different Feature Levels: To investigate the impact of detection at different feature levels, we conduct experiments by directly feeding various feature maps from the RDM into MATLDM, as shown in Table VII. The results indicate that using feature maps from either the RSB or CAB alone yields faster inference times of 5.3 and 4.4 ms, respectively. However, this comes at the cost of reduced recognition accuracy, with AP50 dropping to 78.29% and 77.82%, respectively. Combining both RSB and CAB improves accuracy to an AP50 of 79.26%, while best performance is achieved when all RDM components are utilized. These results demonstrate that feature-level detection can improve inference speed but may compromise accuracy due to incomplete image enhancement.

6) Effect of Different Input Sizes: To evaluate the impact of input image size on the recognition accuracy of the proposed EG-TLR, the experiments are conducted using input sizes of 320×320, 416×416, and 640×640. The results are presented in Table VIII. The findings indicate that recognition performance improves with larger input sizes. At 320×320, the proposed method achieves an AP50 average of 73.93% and an AP50:95 of 32.46%. Increasing the input size to 416×416 significantly boosts performance to an AP50 average of 79.83% and an AP50:95 of 36.62%. This improvement continues with a 640×640 input size, where the model achieves its highest performance, with an AP50 average of 81.80% and an AP50:95 of 39.70%.

VI. CONCLUSION

In this article, we proposed EG-TLR, an enhancement-guided object detection framework for night-time traffic light recognition. EG-TLR comprises the enhancement module RDM and the detection module MATLDM, both trained in an end-to-end manner. The RDM effectively predicts and removes noise from night-time traffic light images, contributing positively to the overall recognition performance. MATLDM enhances the recognition process by learning shadow features through the integration of SDL and MAM, further boosting the framework's effectiveness. Additionally, we construct the Night-TL dataset to advance research on night-time traffic light recognition.

Extensive experiments are conducted to compare EG-TLR with the SOTA methods, demonstrating its effectiveness. With an input size of 416×416, our method achieves an AP50 of 79.83%, significantly outperforming other SOTA methods. Furthermore, this performance is achieved with a processing speed of 5.9 ms and 16.9 GFLOPs, representing an acceptable computational load. The experimental results show that EG-TLR is better at handling challenging night-time traffic light scenarios, such as degraded features, interference from other light sources, and complex environments. Ablation studies further validate the effectiveness of each proposed component. As a part of future work, we plan to explore further optimization of feature-level detection to achieve greater efficiency without compromising accuracy. In addition, night-time traffic light recognition is affected by low-light conditions, color distortion, degraded features, and complex backgrounds, which collectively impact performance. Future work will quantify these factors to further enhance recognition accuracy.

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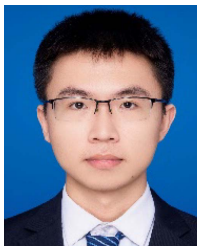
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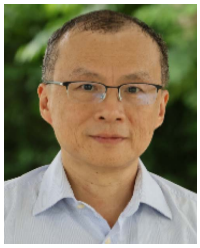
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